Learning Objective 3.5: I can implement beam steering on the ADALM-PHASER and verify the steered pattern against theory.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Building the phase table for a new angle.** Your HB100 locks at 10.525 GHz and the PHASER array has $N = 8$ elements on a $d = 14$ mm pitch. You plan to steer to $\theta_0 = 20°$.

   (a) Find the wavelength and the electrical spacing d/λ.

   $\lambda =$ $\boxed{}$

   $d/\lambda =$ $\boxed{}$

   (b) Compute the progressive element phase $\Delta\phi$ required to steer the beam to 20°.

   $\Delta\phi =$ $\boxed{}$

   (c) Complete the table of phases the beamformer will apply, wrapping each value into 0° to 360°. State which element is the first to wrap.

Element	1	2	3	4	5	6	7	8
Applied phase								

   (d) A classmate says the wrapped table cannot produce the same beam as the unwrapped ramp, because element 7 now leads element 1 by only 2.9° instead of lagging a full turn behind. In one or two sentences, explain why the two tables give identical patterns.

2. **Reading a measured sweep.** A cadet places the HB100 at $+30.0^\circ$ on the protractor arc, hunts for the peak in the Beam Steering box, and then runs a full sweep on the Rectangular tab. The trace peaks at a commanded angle of 28.1°, its half-power crossings are at 20.5° and 35.8°, the peak sits 0.7 dB below the frozen boresight trace, and the first sidelobes read -11.8 dBc. The steer resolution is the default 2.8125°.

   (a) Find the measured half-power beamwidth and compare it with the value predicted from the 13.1° boresight beamwidth.

   θ_{HP} =

   predicted =

   (b) Express the difference between the physical angle and the commanded angle at the peak as a fraction of one sweep grid step.

   offset =

   (c) The cadet reports the array as broken because the beam did not go to 30°. In two or three sentences, say whether the data support that conclusion.

   (d) The peak fell 0.7 dB when the beam was steered to 30°, while the projected-aperture rule predicts a drop of $10\log_{10}(\cos(30^\circ)) = -0.6$ dB instead. Give the most likely reason the measured loss is the larger of the two, and state whether you would expect the gap to grow or shrink at 50°.

3. **Bounding the error sources.** You need to state an uncertainty for the peak angle you measured. Work with the default 2.8125° steer resolution, a source range of 1.00 m, and an HB100 nominal frequency of 10.525 GHz.

   (a) From the sweep grid alone, give the worst-case error in the peak angle read off the trace, and the RMS error if the true peak is equally likely to fall anywhere between two grid points.

   worst case = $\boxed{}$

   RMS = $\boxed{}$

   (b) The phases were computed for 10.525 GHz and applied for $\theta_0 = 45°$. The module then drifts to 10.325 GHz. Find the angle the beam points at, and the resulting error.

   $\theta =$ $\boxed{}$

   $\Delta\theta =$ $\boxed{}$

   (c) How far along the arc must the tripod be misplaced to introduce a 1.0° error in the reference angle?

   $s =$ $\boxed{}$

   (d) Multipath in the classroom adds roughly 1° of peak-angle wander. Combine all four contributions in quadrature, then say which one you could reduce from the GUI and how, and which one you could not.

4. **Working backwards from the phase read-back.** A lab partner recorded the Phase Control values below but forgot which steer angle was commanded. The HB100 was at 10.525 GHz.

Element	1	2	3	4	5	6	7	8
Applied phase	0.0°	125.0°	250.1°	15.1°	140.1°	265.2°	30.2°	155.2°

(a) Recover the progressive phase $\Delta\phi$ from the table.

$\Delta\phi =$

(b) Find the commanded steer angle.

$\theta_0 =$

(c) The same read-back is also consistent with $\Delta\phi = 125.0^\circ + 360^\circ = 485.0^\circ$. Explain in one or two sentences why that second solution can be discarded for this array.

(d) The partner later finds the HB100 was running at 10.300 GHz while these phases were applied. Where did the beam point?

$\theta_0 =$

5. **Two frozen traces.** A group freezes the boresight sweep, then steers to 45° and sweeps again. The boresight trace has a half-power width of 13.1° and defines the 0 dB reference. The steered trace peaks at 45.0°, is 18.9° wide at half power, and its peak is 1.7 dB below the boresight peak.

   (a) Predict the half-power width at 45° and compare it with the measurement.

   $\theta_{\text{HP}} =$ $\boxed{}$

   (b) Predict the peak drop from the projected-aperture rule and compare it with the measured 1.7 dB drop.

   $\Delta G =$ $\boxed{}$

   (c) In one or two sentences, name the single geometric fact behind both the wider beam and the lower peak.

   (d) A radar specification calls for a 13° beam maintained out to 45° of scan. How many elements would this array need at the same 14 mm spacing, and state the cost of that choice in one phrase.

   $N =$ $\boxed{}$

Documentation: ____